



American International University- Bangladesh (AIUB)
Faculty of Engineering (FE)

Course Name:	DATA COMMUNICATION	Course Code:	COE3103
Semester:	Spring 2024-25	Section:	L
Faculty:	DR. AMIRUL ISLAM	Group:	05

Experiment No:	06
Experiment Name:	Study of Digital to Analog Conversion using MATLAB

Submitted by (NAME):	Afnan Bin Islam Nahin	Student ID:	22-49350-3
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Performance Date:	28/4/2025	Due Date:	5/5/2025
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Marking Rubrics (to be filled by Lab Instructor)

Category	Proficient [6]	Good [4]	Acceptable [2]	Unacceptable [1]	Secured Marks
Theoretical Background, Methods & procedures sections	All information, measures and variables are provided and explained.	All Information provided is sufficient, but more explanation is needed.	Most information is correct, but some information may be missing or inaccurate.	Much information is missing and/or inaccurate.	
Results	All of the criteria are met; results are described clearly and accurately;	Most criteria are met, but there may be some lack of clarity and/or incorrect information.	Experimental results don't match exactly with the theoretical values and/or analysis is unclear.	Experimental results are missing or incorrect;	
Discussion	Demonstrates thorough and sophisticated understanding. Conclusions drawn are appropriate for analyses;	Hypotheses are clearly stated, but some concluding statements not supported by data or data not well integrated.	Some hypotheses missing or misstated; conclusions not supported by data.	Conclusions don't match hypotheses, not supported by data; no integration of data from different sources.	
General formatting	Title page, placement of figures and figure captions, and other formatting issues all correct.	Minor errors in formatting.	Major errors and/or missing information.	Not proper style in text.	

Writing & organization	Writing is strong and easy to understand; ideas are fully elaborated and connected; effective transitions between sentences; no typographic, spelling, or grammatical errors.	Writing is clear and easy to understand; ideas are connected; effective transitions between sentences; minor typographic, spelling, or grammatical errors.	Most of the required criteria are met, but some lack of clarity, typographic, spelling, or grammatical errors are present.	Very unclear, many errors.	
Comments:				Total Marks (Out of 30):	

Title: Study of Digital to Analog Conversion using MATLAB

Abstract:

The experiment titled *Study of Digital to Analog Conversion using MATLAB* aims to provide practical insight into how digital data is transformed into analog signals using various modulation techniques. Using MATLAB, students implement and observe modulation methods like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) to encode binary sequences. These techniques demonstrate how the properties of a carrier signal—such as its amplitude, frequency, or phase—can be adjusted to represent digital information. The experiment helps students visualize signal behavior in the time domain and better understand digital-to-analog conversion processes. Overall, the activity enhances comprehension of signal modulation and supports learning in digital communication and signal processing.

Introduction:

Digital-to-Analog Conversion (DAC) plays a crucial role in communication systems by converting digital signals into analog waveforms that can be transmitted over physical media. This process allows binary data to be represented using continuous analog signals. Techniques like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) are commonly used to modulate a carrier signal by altering its amplitude, frequency, or phase based on the digital input. In this experiment, MATLAB is utilized to simulate and analyze these modulation schemes, enabling students to observe how digital information is transformed into analog form. The exercise helps build a solid foundation in signal processing and modulation by providing hands-on experience with digital-to-analog signal conversion and visualization.

Performance Task:

My ID: 22-49350-3

A=2
B=2
C=4
D=9
E=3
F=5
G=0
H=3

Here 8-bit ASCII Code of the following values are:

Symbol	E	F	G
Decimal Value	3	5	0
8-bit ASCII Code	00110011	00110101	00110000

(1) 8-ASK: Different amplitudes for 000 to 111 in the modulated signal can be 0 V, 1 V, 2 V, 3 V, 4 V, 5 V, 6 V, and 7 V respectively.

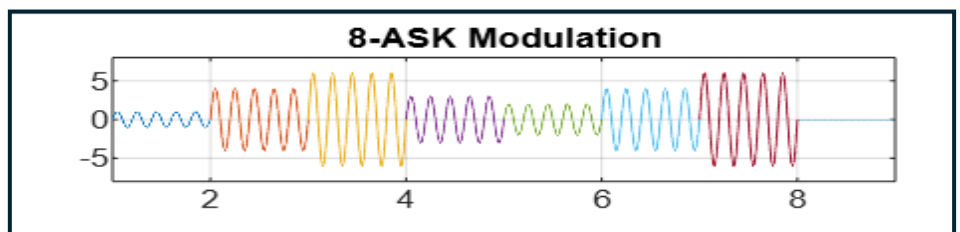
```
close all;  
clc;  
f = 5;
```

```
x = [0 0 1 1 0 0 1 1, 0 0 1 1 0 1 0 1, 0 0 1 1 0 0 0 0];  
symbols = reshape(x, 3, []);  
amplitudes = symbols(:,1)*4 + symbols(:,2)*2 + symbols(:,3);
```

```
nx = length(amplitudes);
```

```
i = 1;  
while i < nx+1  
    t = i:0.001:i+1;  
    A = amplitudes(i);  
    ask = A * sin(2*pi*f*t);
```

```
subplot(3,1,1);  
plot(t, ask);  
hold on;  
grid on;  
axis([1 nx+1 -8 8]);  
title('8-ASK Modulation');
```



```

    i = i + 1;
end

```

(2) 8-FSK: Different frequencies for 000 to 111 in the modulated signal can be 1 Hz, 2 Hz, 3 Hz, 4 Hz, 5 Hz, 6 Hz, 7 Hz, and 8 Hz respectively.

```

close all;
clc;

```

```

Fs = 1000;
f_base = 1;
symbol_duration = 1;

```

```

x = ['3', '5', '0'];
bin_stream = reshape(dec2bin(x, 8).-'0', 1, []);

```

```

nx = length(bin_stream)/3;
i = 1;

```

```

while i <= nx
    idx = (i-1)*3 + 1;
    bits = bin_stream(idx:idx+2);
    symbol = bin2dec(num2str(bits));
    freq = f_base + symbol; % 1Hz to 8Hz

```

```

    t = i:1/Fs:i+symbol_duration-1/Fs;

```

```

    fsk = sin(2*pi*freq*t);

```

```

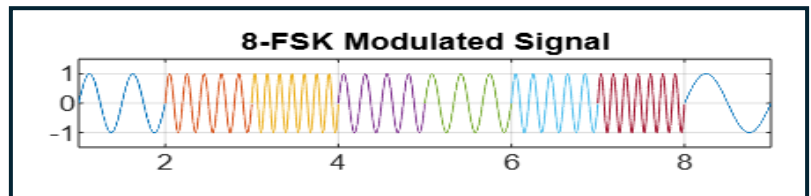
    subplot(3,1,2);
    plot(t, fsk);
    hold on;
    grid on;
    axis([1 nx+1 -1.5 1.5]);
    title('8-FSK Modulated Signal');

```

```

    i = i + 1;
end

```



(3) 8-PSK: Different phases for 000 to 111 in the modulated signal can be 0 , $\pi/4$, $3\pi/4$, $\pi/2$, $-\pi/4$, $-\pi/2$, π , $-3\pi/4$ respectively.

```
close all;
clc;

f = 5;
x = [0 0 1 1 0 0 1 1, 0 0 1 1 0 1 0 1, 0 0 1 1 0 0 0 0];

nx = length(x);

phases = [0, pi/4, 3*pi/4, pi/2, -pi/4, -pi/2, pi, -3*pi/4];

i = 1;
while i <= nx
    bitgroup = x(i:i+2);

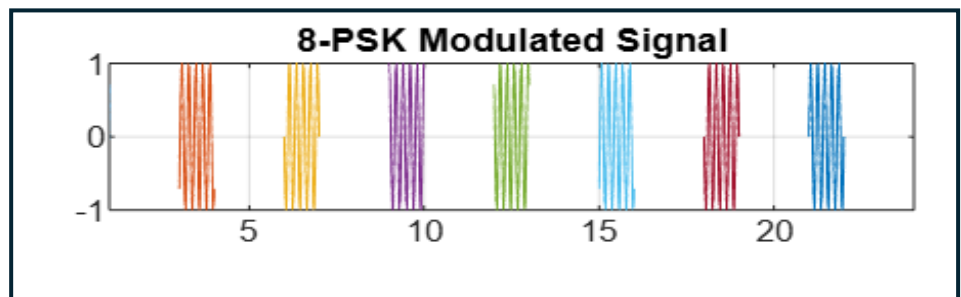
    phase_index = bitgroup(1)*4 + bitgroup(2)*2 + bitgroup(3)*1 + 1;

    phase = phases(phase_index);

    t = (i-1):0.001:i;
    psk = sin(2*pi*f*t + phase);

    subplot(3, 1, 1);
    plot(t, psk);
    hold on;
    grid on;
    axis([1 nx -1 1]);
    title('8-PSK Modulated Signal');

    i = i + 3;
end
```



Result:

In this experiment, we successfully demonstrated the process of converting a digital bitstream into analog signals using three modulation techniques: 8-ASK (Amplitude Shift Keying), 8-FSK (Frequency Shift Keying), and 8-PSK (Phase Shift Keying).

For 8-ASK, the bitstream was divided into 3-bit groups, and each group was mapped to a unique amplitude level from 0V to 7V. The resulting analog waveform displayed distinct amplitude levels corresponding to each symbol.

For 8-FSK, each 3-bit symbol was assigned a unique frequency ranging from 1 Hz to 8 Hz. The output analog signal showed clear frequency variations that matched the expected mapping from binary input to frequency.

For 8-PSK, we used 3-bit symbols to represent eight different phase shifts. The generated analog signal exhibited periodic sine waves with different phase shifts such as 0 , $\pi/4$, $3\pi/4$, and so on.

This clearly illustrated how phase can be modulated to carry digital information.

The time duration for each symbol was maintained at 1 second, ensuring consistent symbol spacing in all three modulation schemes. The plots obtained from MATLAB confirmed that each digital-to-analog conversion method was implemented correctly and visualized effectively.

Discussion:

This experiment demonstrated how digital data can be converted into analog signals using 8-ASK, 8-FSK, and 8-PSK modulation techniques in MATLAB. In 8-ASK, data was represented by varying amplitudes, though it's more prone to noise. 8-FSK used different frequencies, offering better noise resistance. 8-PSK encoded data by changing signal phases, making it bandwidth-efficient but sensitive to phase errors. The MATLAB results confirmed successful implementation of each technique, highlighting their respective strengths and limitations.

Conclusion:

The experiment successfully demonstrated digital-to-analog conversion using 8-ASK, 8-FSK, and 8-PSK modulation in MATLAB. Each method effectively encoded a 24-bit stream into analog signals, showcasing their practical application and differences in noise sensitivity, bandwidth usage, and implementation complexity. This enhances understanding of how digital data is transmitted over analog channels.

References:

- Prakash C. Gupta, *"Data communications"*, Prentice Hall India Pvt.
- William Stallings, *"Data and Computer Communications"*, Pearson
- Forouzan, B. A. *"Data Communication and Networking"*, Tata McGraw. (2005).
- AIUB Data Communication Engineering Lab Manual, Report 06